

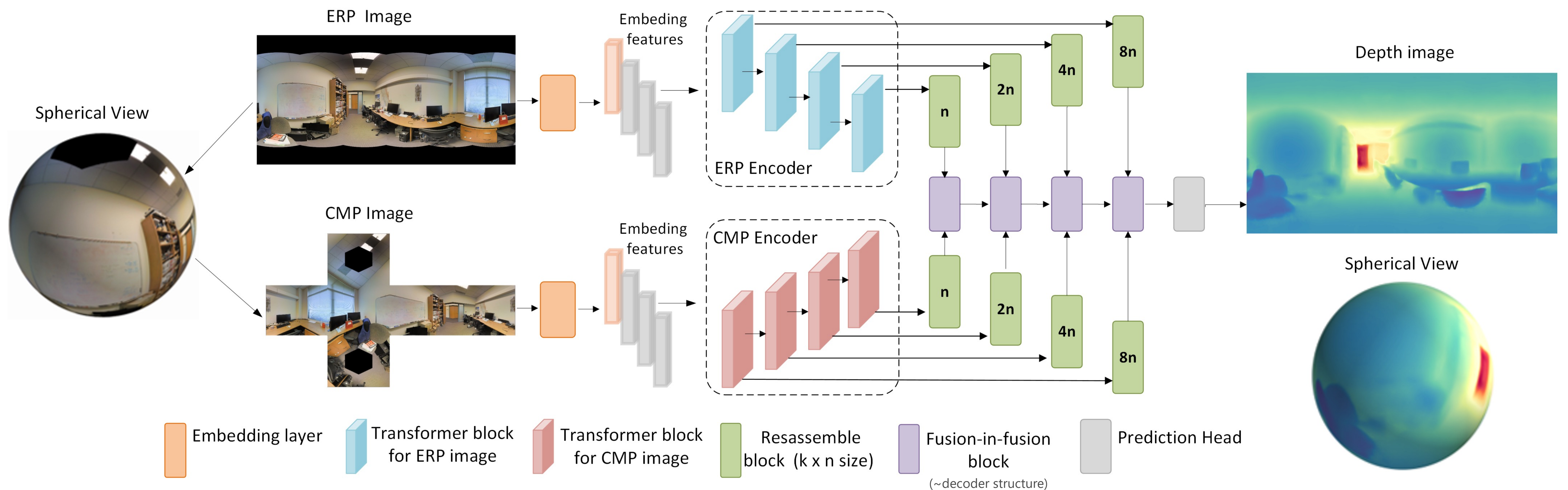
FuseDPT: Multi-scale and Multi-Projection Model for Learning Depth in 360 Degree

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Motivation and Background:

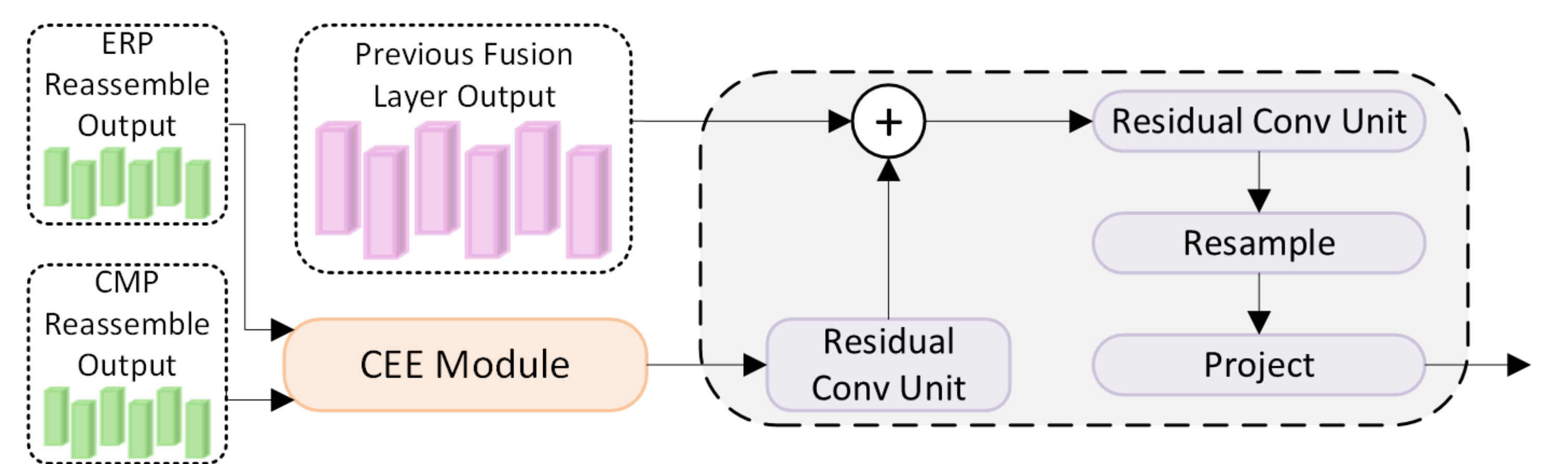
- 360° cameras: complete scene perception
- Applications: robotics, localization, immersive media
- Challenge: estimating depth from omnidirectional images due to distortions of the equirectangular projection (ERP)
- Related and base work:

Depth Anything [1]: DPT (Depth Prediction **T**ransformer) [2] pretrained on large-scale datasets of perspective images. Strong foundation for feature extraction but no handling of omnidirectional distortions.

UniFuse [3]: CNN (Convolutional Neural Network)-based encoder-decoder that employs feature **fusion** between ERP and Cube Map **Projections** (CMP) to improve omnidirectional depth.

Proposed FuseDPT contributions:

- Dual-projection **transformer**: ERP and CMP representations as input
- **Fusion-in-fusion**: Multi-projection **feature fusion** into **DPT's fusion block**
→ Multi-projection and multi-scale feature integration in one fusion module.



CEE: Cubemap-to-Equirectangular Enhancement module [3]

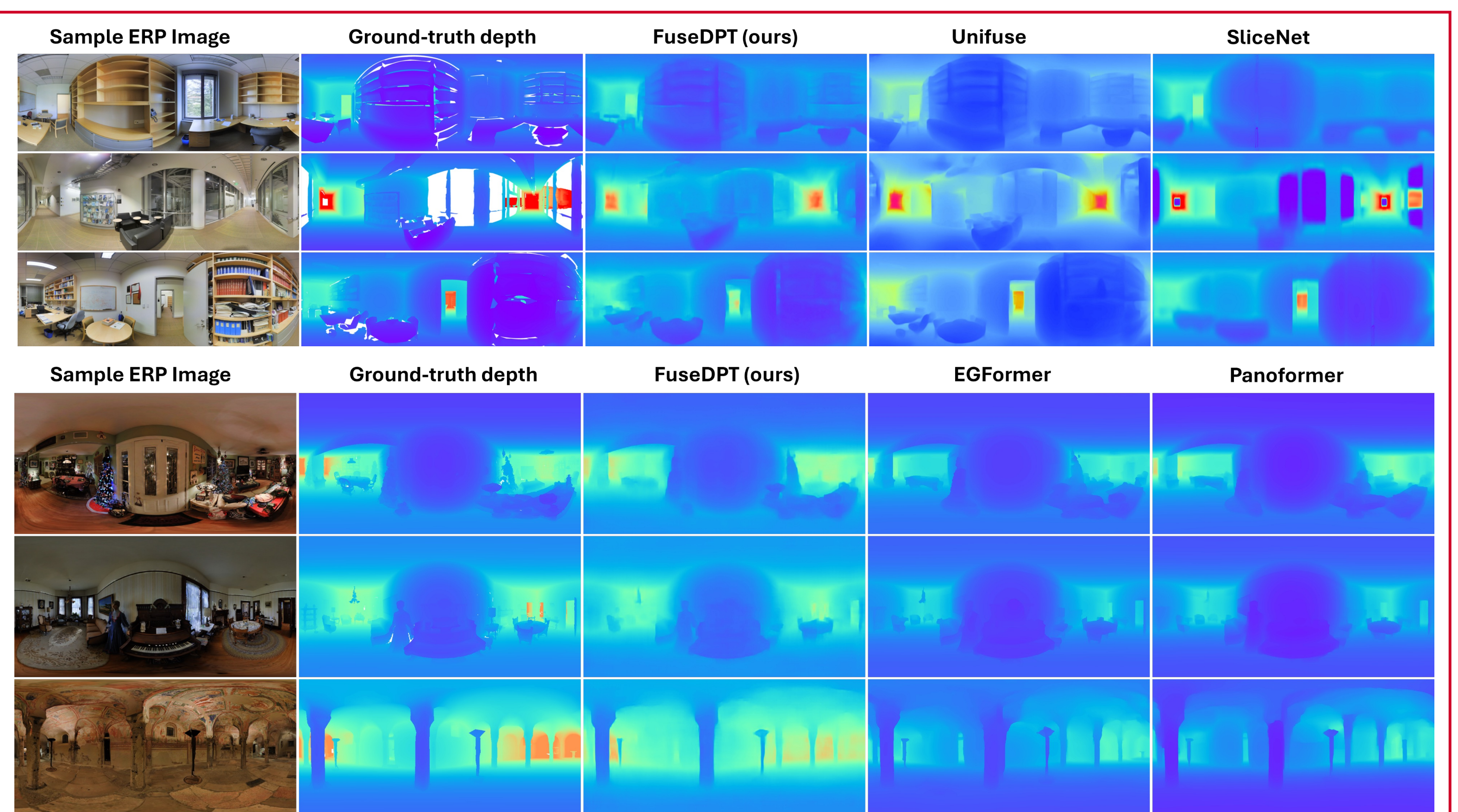
Results:

Stanford2D3D [4] benchmark

Models	$\delta_{1.25}^1 \uparrow$	$\delta_{1.25}^2 \uparrow$	$\delta_{1.25}^3 \uparrow$	Abs. Rel. $\downarrow$	RMSE $\downarrow$
UniFuse [12]	0.871	0.966	0.988	0.111	0.369
SliceNet [22]	0.903	0.963	0.984	0.176	0.350
HoHoNet [29]	0.905	0.969	0.989	0.101	0.383
FreDSNet [4]	0.842	0.958	0.986	-	0.273
S ² Net [17]	0.919	0.978	0.991	0.090	0.338
Panda-S [6]	0.953	0.986	0.994	0.076	0.287
Panda-L [6]	0.968	0.991	0.995	0.061	0.254
FuseDPT (ours)	0.972	0.993	0.997	0.053	0.229

Pano3D [5] benchmark

Models	$\delta_{1.25}^1 \uparrow$	$\delta_{1.25}^2 \uparrow$	$\delta_{1.25}^3 \uparrow$	Abs. Rel. $\downarrow$	RMSE $\downarrow$	#params	FLOPS
BiFuse[30]	0.768	0.925	0.973	0.170	0.727	253.0M	723.4G
SliceNet [22]	0.857	0.964	0.987	0.125	0.576	79.5M	84.3G
Yun et al. [35]	0.913	0.979	0.992	0.091	0.470	123.7M	589.4G
Panoformer [27]	0.944	0.985	0.994	0.0699	0.4046	20.4M	77.7G
EGFormer [36]	0.950	0.988	0.995	0.034	0.276	15.4M	73.9G
FuseDPT (ours)	0.914	0.986	0.995	0.099	0.336	46.3M	157.0G



Key Findings & Limitations:

- ✓ **Transfer learning** from Depth Anything [1] **generalizes** to **panoramic depth** estimation
- ✓ **Fine-tuning** pretrained **encoders** further **improves** ERP/CMP **depth** prediction **accuracy**
- ✓ **Data augmentation** significantly enhances **robustness** and **generalization**
- ✓ Training with **small-batch size** **performs better** than larger batch sizes in our experiments
- ✓ **FuseDPT's simple** and **generic architecture** achieves **superior** results on **Stanford2D3D** [4]

X Evaluation was limited to the ViT-S backbone due to hardware constraints.

X Current inference speed (~4–5 FPS) can hinder some applications.

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References

1. Yang et. al., IEEE/CVF CVPR Workshops 2024
2. Ranftl et. al., IEEE/CVF CVPR Workshops 2021
3. Jiang et. al., IEEE Robotics and Automation Letters (RAL) 2021
4. Armeni et al., arXiv/1702.01105 2019
5. Albanis et. al., IEEE/CVF CVPR Workshops 2021